

Computational Thinking

Week-8 Summary

Dictionaries with lists as values(Lecture 8.1) followup lecture to 7.1:

Example1:

City by card no's as list

Dictionary: city by card
Chennai : [2, 10, 17...]
Banglore : [9, 23....]
.....

Example2:

Shop with customers as list

Dictionary: shop with customer
Sun General : [Abhinav,]
SV Stores : [Advait, Ahmed, Abhinav, Srivatsav...]
.....

Note: Using above list you can compute some commonality between stores and customers

Resolve pronoun with its equivalent matching noun using dictionary and ordered list (Lecture 8.2):

Create ordered list of pronouns and nouns as two different dictionaries

{ Pronoun : [0, 9, 11, 27....]}
{ Noun : [2, 3, 4, 10.....]}

Using above dictionaries you can find nearest noun to pronoun [9](his) is noun[4](Swaminathan), which is occurred before pronoun. And pronoun[11](he) is refering earlier noun[4](Swaminathan).

Dictionary comparison to find common elements in them(Lecture 8.3):

(Using Train Dataset)

Example1:

First start from Mangalore to a junction ?(X) and from that junction go to Pune. Is it possible?

Mangalore -> ? -> Pune

- First get each train from Mangalore station. For each train find the list of all stations that the train goes

Mangalore -
02283 [Madgaun, Ratnagiri, Mumbai...
02284]
12223 [.....
12224]

Pune -
12219 [Secunderabad, Solapur, Mumbai
12220]

12221[.....
12222]

Here you can reach a station/junction Mumbai through 02283, from there you can reach Pune through 12219.

Procedure to find relations among customers based on their spending patterns(Lecture 8.4):

Here each card is processed by separate procedure and returns sum of the items in each category. The returned list is used by other pseudocode

Name	A	F	T	U	H
Advait	0	0+3	0	4	0
Abhinav	0	0	2	0	0
.....					
...					

- Every time new customer is created in dictionary. Add to category sum, if customer is existed in dictionary

- Here we should able take card of the sub procedure and its return value. The side effects is the problem when you write sub-procedures.

Side effects in pseudocodes for lists and dictionaries(Lecture 8.5):

Comparing two lists for duplicate items(Unsorted lists):

```
Procedure FindOverlap(l1,l2)
    overlap = []
    foreach x in l1 {
        foreach y in l2 {
            if (x == y) {
                overlap = overlap ++ [x]
            }
        }
    }
    return(overlap)
End FindOverlap
```

Comparing two lists for duplicate items(Sorted lists):

- Need not start inner loop from beginning
- Take the help of first() and rest() functions to cutdown second list

```

Procedure FindOverlap2(l1,l2)
    overlap = []
    if (length(l2) == 0) {
        return(overlap)
    }
    else {
        y = first(l2)
        l2 = rest(l2)
    }
    foreach x in l1 {
        while (y < x and length(l2) > 0) {
            y = first(l2)
            l2 = rest(l2)
        }
        if (x == y) {
            overlap = overlap ++ [x]
        }
    }
    return(overlap)
End FindOverlap2

```

Note: Here in the above procedure list l2 is modified. So it's side effect. To avoid this create a local copy of it.

```

Procedure FindOverlap3(l1,l2)
    overlap = []
    if (length(l2) == 0) {
        return(overlap)
    }
    else {
        myl2 = l2
        y = first(myl2)
        myl2 = rest(myl2)
    }
    foreach x in l1 {
        while (y < x and length(myl2) > 0) {
            y = first(myl2)
            myl2 = rest(myl2)
        }
    }
    return(overlap)
End FindOverlap3

```

```

        }
        if (x == y) {
            overlap = overlap ++ [x]
        }
    }
    return(overlap)
End FindOverlap3

```

Sometimes side effect necessary:

Example1:
Deleting a key from dictionary

```

Procedure DeleteKey(d,k)
    myd = {}
    foreach key in keys(d) {
        if (k != key) {
            myd[key] = d[key]
        }
    }
    d = myd
End DeleteKey

```

Example2:
Deleting a key from dictionary without side effect:

```

Procedure DeleteKey2(d,k)
    myd = {}
    foreach key in keys(d) {
        if (k != key) {
            myd[key] = d[key]
        }
    }
    return(myd)
End DeleteKey2

```

```

myd = DeleteKey2(myd,k)

```